

PRACTICAL ACTIVITY 4 [CHALLENGE 4]: b00tl3gRSA3 Write-up

b00tl3gRSA3

Hard

Cryptography

picoCTF 2019

AUTHOR: INVISIBILITY

Description

Why use p and q when I can use more?

Connect with nc fickle-tempest.picoctf.net 58618.

This challenge launches an instance on demand.

Its current status is: RUNNING

Instance Time Remaining: 29:53

Restart Instance

Hints

1

There's more prime factors than p and q, finding d is going to be different.

1,778 users solved

95% Liked

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CHALLENGE TITLE: **b00tl3gRSA3**
(<https://play.picoctf.org/practice/challenge/78?category=2&page=1&search=b00tl3gRSA3>)
PLATFORM: **picoCTF**
AUTHOR: **invisibility**
CATEGORY: **Cryptography (Hard)**
TOOL USED: **Kali Linux**

Challenge Overview:

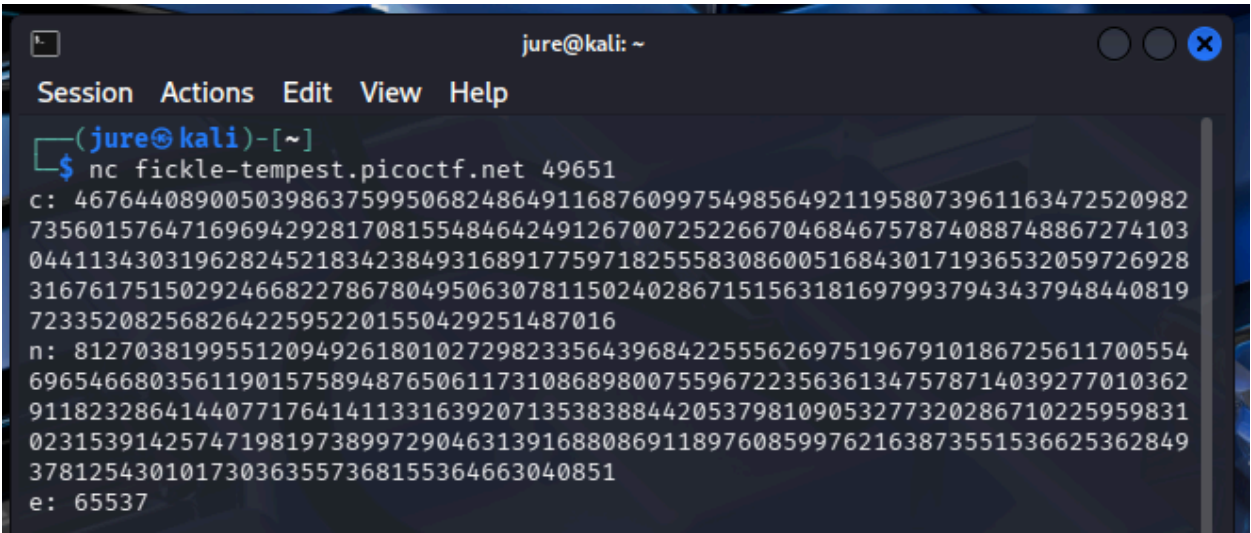
The **b00tl3gRSA3** challenge from picoCTF is a hard cryptography problem that explores a variation of RSA where the modulus **n** is generated from more than two prime numbers. The challenge description hints at this by stating, “Why use p and q when I could use more,” and that finding the private key **d** will be different. In standard RSA, the modulus is calculated as:

$$n = p \times q,$$

where **p** and **q** are two large primes. However, in this challenge the modulus is composed of multiple prime factors instead of just two. Because of this, the usual process for computing the Euler totient and deriving the private exponent **d** requires a slightly different approach, making the challenge more complex than a typical RSA decryption task.

PROCESS:

To begin solving the challenge, I connected to the instance using the terminal in Kali Linux through a netcat connection. Once connected, the server provided the RSA parameters: the ciphertext **c**, the modulus **n**, and the public exponent **e**.



Initially, I attempted to decrypt the ciphertext using an RSA decoding tool by inputting the values for **c**, **n**, and **e**, but the result produced unreadable output.



This indicated that the missing component required to decrypt the message correctly was the private exponent **d**. To compute **d**, I needed the Euler totient value $\phi(n)$. Since the hint suggested that the modulus contained more than two primes, I first factored the value of **n** using an online integer factorization tool.

Integer factorization calculator

alpertron.com.ar/ECM.HTM

Integer factorization calculator

Value176414113316392071353838844205379810905327732028671022595983102315391425747198197389972904631391688086911897608599762163873551536625362849378125430101730363557368155364663040851

Actions

Only evaluateIs prime?FactorHelpConfigOpen wizardFrom fileBlocky modeClear input

Functions

Category: Basic Math

() ^ + - * / % ^ ans sqrt iroot Random Abs Sign

Type one numerical expression or loop per line. Example: x=3;x=n(x);c<=100;x-1

Press the **Help** button to get help about this application. Press it again to return to this screen. Keyboard users can press CTRL+ENTER to start calculation. This is the WebAssembly version.

• 81 270381 995512 094926 180102 729823 356439 684225 556269 751967 910186 725611 700554 696546 680356 119015 758948 765061 173108 689800 755967 223563 613475 787140 392770 103629 118232 864144 077176 414113 316392 071353 838844 205379 810905 327732 028671 022595 983102 315391 425747 198197 389972 904631 391688 086911 897608 599762 163873 551536 625362 849378 125430 101730 363557 368155 364663 040851 (344 digits) = 9016 592927 × 9610 687397 × 9714 043621 × 9727 159781 × 9772 901461 × 10006 806617 × 10119 629879 × 10369 949173 × 10748 696983 × 10845 158113 × 11295 019693 × 11761 768013 × 12152 659411 × 12493 075303 × 12730 966117 × 13008 421207 × 13778 589679 × 13781 204707 × 13944 552137 × 14270 537549 × 14503 233469 × 14803 894367 × 14990 418959 × 15241 510873 × 15556 318727 × 15624 736331 × 16047 612479 × 16120 816939 × 16275 910907 × 16526 569727 × 16528 074239 × 16613 743907 × 16898 049059 × 17073 408617

Number of divisors: 17179 869184

Sum of divisors: 81 270382 211978 204154 775215 089816 766712 805453 254791 766156 497899 763116 467876 556465 214636 559927 374173 835654 274216 121552 883496 702331 956959 975846 260084 359149 261081 205549 696080 171194 410321 642700 249872 455437 158262 623430 992889 378009 380050 644056 009837 662371 839352 272079 057042 950098 747792 674633 429178 305393 932523 833200 358719 677726 797928 155054 080000 000000 (344 digits)

Euler's totient: 81 270381 779045 986256 467354 066027 661820 845063 994674 472852 151436 231226 719600 908777 310846 916561 752120 255421 470451 118724 897890 817729 915384 042284 074377 664641 265145 264730 165029 197671 407291 168083 296949 458258 812171 824993 042558 500593 337861 959501 545926 050701 518435 035109 118789 214804 664023 112740 559283 469612 916241 506899 019715 898087 904885 312532 460339 200000 (344 digits)

Möbius: 1

$$n = a^2 + b^2 + c^2 + d^2$$
$$a = 7393\ 614782\ 054690\ 999036\ 094790\ 564919\ 954679\ 477985\ 103878\ 274055\ 820155\ 568621\ 899676\ 098933\ 834002\ 059054\ 082337\ 624066\ 072287\ 642645\ 869550\ 806064\ 340551\ 592746\ 751698\ 133045$$

26°C Mostly cloudy

Search

ENG US 9:15 PM 3/13/2026

Once **n** was factored into multiple prime numbers, I calculated the totient by multiplying $(p_1-1)(p_2-1)(p_3-1)\dots$ for all prime factors of **n**. With the totient value computed, I then calculated the private exponent **d** as the modular inverse of **e modulo $\phi(n)$** . After obtaining **d**, I used it to decrypt the ciphertext using the RSA decryption formula $m = c^d \bmod n$.

RSA Cipher Calculator - Online

dcode.fr/rsa-cipher

Search for a tool

Search a tool on DCode

Browse the full DCode tools list

Results

GoDaddy

RSA CIPHER

Cryptography - Modern Cryptography - RSA Cipher

RSA DECODER

Indicate known numbers, leave remaining cells empty.

VALUE OF THE CIPHER MESSAGE (INTEGER) C=

PUBLIC KEY E (USUALLY E=65537) E=

PUBLIC KEY VALUE (INTEGER) N=

PRIVATE KEY VALUE (INTEGER) D=

FACTOR 1 (PRIME NUMBER) P=

FACTOR 2 (PRIME NUMBER) Q=

INTERMEDIATE VALUE PHI (INTEGER) phi=

Summary

RSA Decoder

RSA Certificate Reader

Complementary Helper tools

What is the RSA cipher? (Definition)

How to encrypt using RSA cipher?

How to decrypt a RSA cipher?

How to generate RSA keys?

How to recognize RSA ciphertext?

What are possible RSA attacks?

How to decrypt RSA without the private key?

The resulting value was then converted into readable text, which revealed the flag and successfully solved the challenge.

b00tl3gRSA3

Hard

Cryptography

picoCTF 2019

AUTHOR: INVISIBILITY

Description

Why use p and q when I can use more?

Connect with nc fickle-tempest.picoctf.net 49651.

This challenge launches an instance on demand.

Its current status is: **RUNNING**

Instance Time Remaining: **19:30**

Restart Instance

Hints

1

1,779 users solved

95% Liked

picoCTF(too_many_fact0rs_3023548)

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Hurray! You solved this challenge.

Reflection:

I learned that RSA can technically use more than two prime factors in its modulus, but doing so changes how the totient and private key must be calculated. The challenge also reinforced the importance of understanding the mathematical structure behind RSA, especially how factoring the modulus allows the recovery of the private key. Overall, the exercise improved my ability to analyze RSA parameters and apply the correct formulas when the implementation differs from the standard case.